

Neuroscience Assignment - Group 02

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1. Introduction

The aim of this assignment was to investigate how closely different auditory models represent sounds compared with the human brain. We used responses to 288 sounds from the superior temporal gyrus (STG) and compared them with three supplied models: WaveformModel, UninspiredModel and InspiredModel, as well as YAMNet. Using representational similarity analysis (RSA), we examined the effects of training, model architecture and layer depth on brain alignment. This is mainly a contrastive study because existing models and their representations were compared against the same brain data. Category visualisations were also used to support the interpretation.

2. Methods

2.1. RDM construction

We loaded STG responses to 288 sounds from one participant using SantoroDataset and constructed a brain representational dissimilarity matrix (RDM) using correlation distance. For each of the three architectures, we extracted activations from every trainable layer for one untrained model and five trained checkpoints. Layer outputs were flattened into one feature vector per sound and converted into RDMs using the same distance metric. YAMNet activations were loaded and processed in the same way. Sound order was kept identical across all representations, resulting in 288×288 RDMs.

2.2. RSA comparison and statistical analysis

We compared each model RDM with the brain RDM using the Spearman correlation between their upper triangles (41,328 sound pairs). We call this the brain alignment. We also repeated the main comparison with Pearson correlation and Kendall's tau to check that the results did not depend on the metric.

The five trained runs of each model were kept separate. For each layer, we report the mean over the runs with a 95% confidence interval based on the t-distribution. The untrained models and YAMNet were single models, so they have no confidence interval. Each model was summarised by the best layer of each run.

We compared trained and untrained models with a one-sample t-test, and the three architectures in pairs with Welch's t-tests and a Bonferroni correction. To test the effect of depth, we correlated layer position with alignment for each run. We used a significance level of 0.05.

2.3. t-SNE visualisation

We used t-SNE to visualise the sound representations in two dimensions. We compared the brain responses with the first layer and classifier of each custom model, using trained run 0. For YAMNet, we selected the first layer and layer12relu, which had the highest RSA score. We used perplexity 30 and random state 42 for all plots. Sounds were coloured by six categories taken from their filenames: animal, music, nature, speech, tools and voice. We compared the category groupings within each plot, rather than the positions across separate plots.

3. Results

3.1. Effect of training

Training made the middle layers more brain-like. Most layers of the waveform model improved (up to +0.066). The convolutional layers in the uninspired model increased (+0.006 to +0.052), as well as the second and third convolutional layers in the inspired model (+0.055 and +0.073). In the late layers, training did not help: the fully connected layers of the uninspired model became less brain-like (−0.028 to −0.034), and the RNN layer of the inspired model changed only slightly (−0.011).

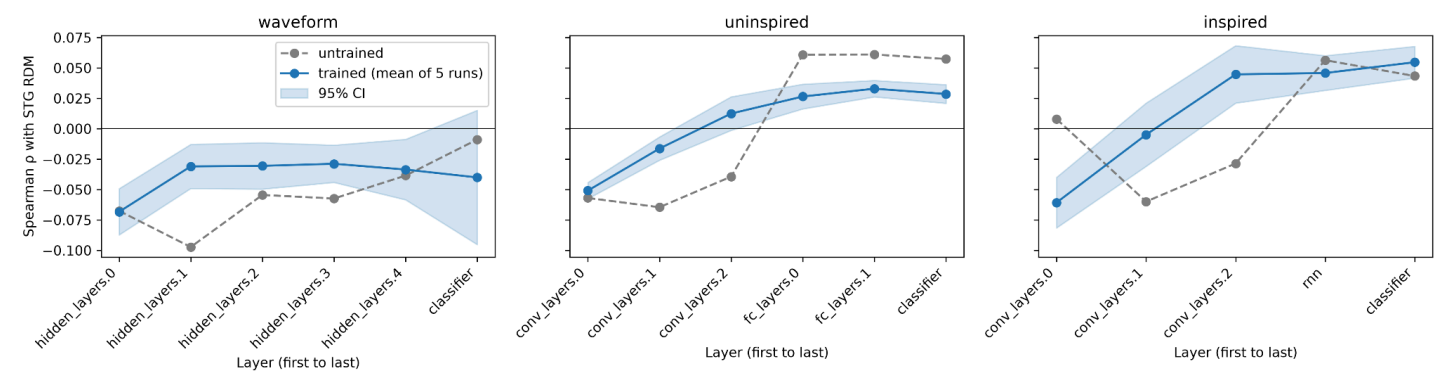


Figure 1. *Brain alignment (Spearman ρ between model and STG RDMS) per layer for the untrained models (grey, dashed) and the trained models (blue, mean of five runs; shaded band = 95% confidence interval). Where the blue line lies above the grey line, training made that layer more brain-like.*

In the table below, we compare the best-performing layer of each untrained model with the mean of the five trained runs. We can see that training did not increase the maximum alignment of any model. For the uninspired model, alignment decreased a lot after training ($\rho = 0.061$ vs 0.033 , corrected $p = 0.001$), and for the inspired model the difference was not significant ($\rho = 0.056$ vs 0.058 , corrected $p = 1.00$). The waveform model also showed no significant difference ($\rho = -0.009$ vs -0.019 , corrected $p = 0.74$), and its 95% CI included zero, meaning there was no reliable match with the brain data. Together with the results shown in Figure 1, these data imply that training primarily enhances the level of alignment in the middle layers, while the untrained models already achieve comparable or higher alignment in their late layers.

A limitation is that each untrained model is a single random initialisation, so the untrained baseline has no confidence interval and could differ with another random start.

Table 1. Best-layer alignment of the untrained model and the five trained runs, compared with a one-sample t-test (Bonferroni corrected for three tests).

Model	Untrained	Trained (mean)	95% CI	p (corrected)
Waveform	−0.009	−0.019	[−0.040, 0.002]	0.742
Uninspired	0.061	0.033	[0.026, 0.040]	0.001
Inspired	0.056	0.058	[0.044, 0.071]	1.000

3.2. Effect of architecture

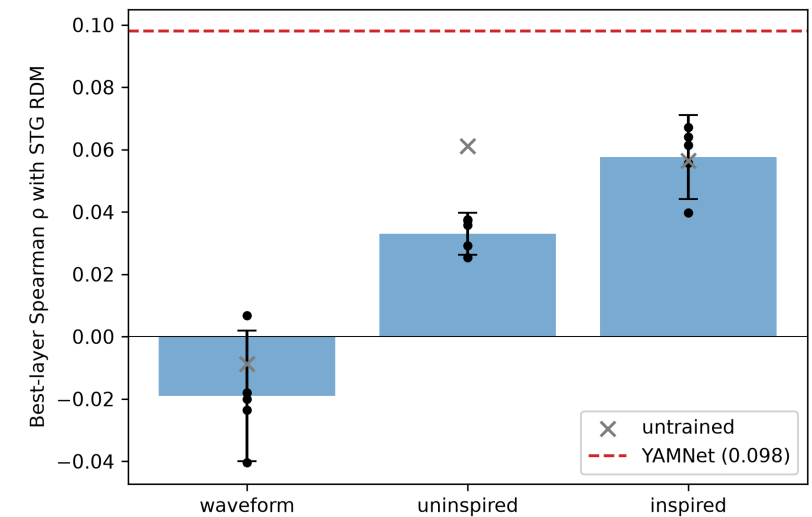


Figure 2. Best-layer brain alignment per architecture. Bars show the mean of five trained runs with 95% confidence intervals, black dots the individual runs, grey crosses the untrained models, and the red dashed line YAMNet.

The best-layer alignment of the models can be seen in Figure 2. As the steps were added, the alignment increased. The waveform model has a poor alignment with the brain:its score was around zero ($\rho = -0.019$, 95% CI $[-0.040, 0.002]$). The uninspired model scored higher ($\rho = 0.033$ $[0.026, 0.040]$), and the inspired model scored highest ($\rho = 0.058$ $[0.044, 0.071]$). All three models were significantly different from each other (Table 2).

The biggest improvement came from using a spectrogram as input and adding convolutional layers (waveform vs uninspired). Adding a recurrent layer (GRU) and ReLU6 improved alignment a bit more (uninspired vs inspired). Interestingly, the waveform model has the most parameters but scored lowest, so the way a model is built matters more than its size. In fact, even before training, the convolutional models already outperform the waveform model (uninspired 0.061 and inspired 0.056 vs waveform -0.009). Compared with our models, YAMNet, a larger model trained on much more audio data, achieved even better results ($\rho = 0.098$).

Table 2. Pairwise comparison of the trained architectures (Welch's t-test, Bonferroni corrected).

Comparison	Design step	Difference	p (corrected)
Waveform vs uninspired	+ spectrogram, convolutions	-0.052	0.004
Waveform vs inspired	+ spectrogram, convolutions, GRU, ReLU6	-0.077	< 0.001
Uninspired vs inspired	+ GRU, ReLU6	-0.025	0.012

3.3. Effect of layer depth

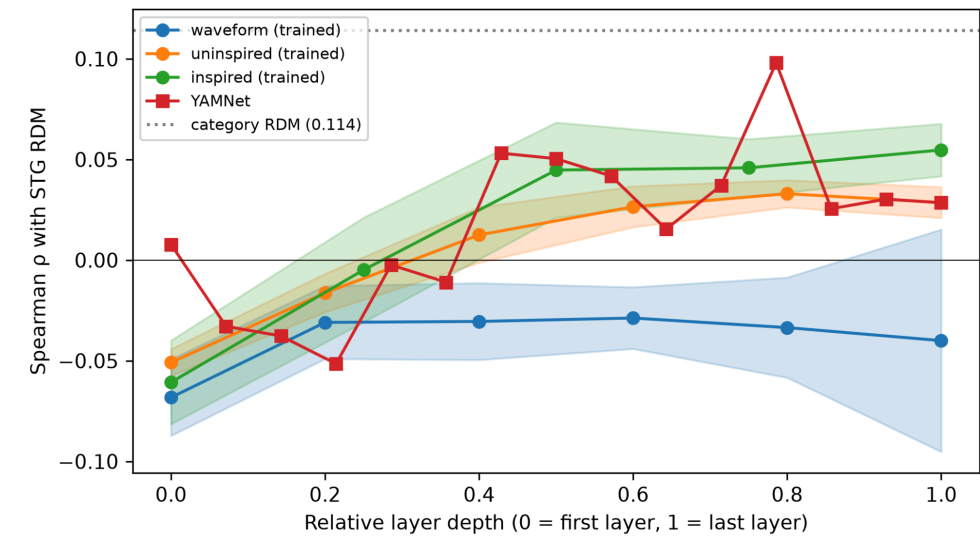


Figure 3. Brain alignment by relative layer depth (0 = first layer, 1 = last layer) for the trained models (mean of five runs with 95% CI) and YAMNet. The dotted line shows the score of the simple category RDM.

In the third graph we plot how brain-like each layer (from first (0) to last (1)) is. The uninspired and inspired models show an increase in brain-likeness as the network deepens: the first layers fall below zero and the last layers reach the highest values. This was very consistent over five runs (uninspired: $\rho = 0.89$, 95% CI $[0.81, 0.96]$, inspired: $\rho = 0.90$, 95% CI $[0.75, 1.05]$, both corrected $p < 0.001$). The waveform model shows no consistent trend ($\rho = 0.31$, 95% CI $[-0.39, 1.02]$, corrected $p = 1.00$) and remained below zero across all layers.

YAMNet was also somewhat more brain-like in the deeper layers of the network, but not monotonically and in a way that was not significant after correction ($\rho = 0.61$, corrected $p = 0.062$). It scored low in the early layers, reached the maximum at layer 12 ($\rho = 0.098$), and then dropped again in the final layers. In this network, the most brain-like layer was somewhere deep in the network, but not the last one.

3.4. Different RSA metrics

Here we want to check that our results do not depend on how we compare the RDMs, so we repeated the best-layer comparison with two other metrics: Pearson correlation and Kendall's tau (Table 3). The order of the models stayed the same for all three metrics: YAMNet scored highest, then the inspired, uninspired and waveform models. Kendall's tau gave smaller numbers overall, but this is normal for this metric. So our conclusions about architecture hold no matter which metric we use.

Table 3. Best-layer alignment for three RSA metrics (trained models: mean of five runs with 95% CI).

Model	Spearman	Pearson	Kendall
Waveform	-0.019 [-0.040, 0.002]	-0.022 [-0.041, -0.003]	-0.013 [-0.027, 0.001]
Uninspired	0.033 [0.026, 0.040]	0.031 [0.022, 0.039]	0.022 [0.017, 0.026]
Inspired	0.058 [0.044, 0.071]	0.054 [0.034, 0.074]	0.038 [0.029, 0.047]
YAMNet	0.098	0.077	0.063

3.5. Category visualisations

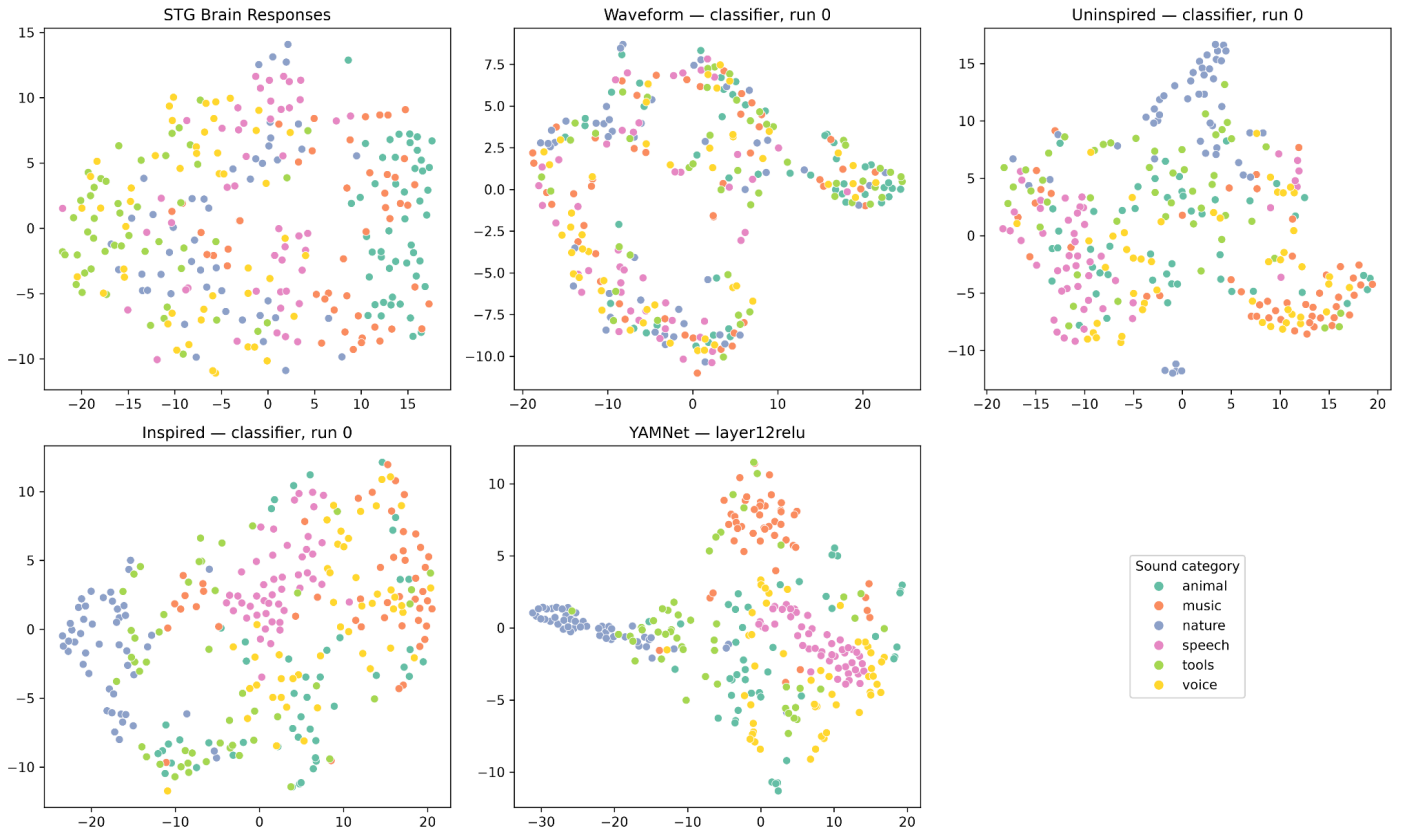


Figure 4. t-SNE visualisations of STG brain responses, the classifier layers of the three custom models (trained run 0), and YAMNet's layer12relu. Each point represents one sound, coloured by its broad category.

Figure 4 shows overlapping categories in the STG responses, with some broad grouping. The waveform classifier remains mostly mixed, while the uninspired and inspired models show clearer grouping of nature sounds, with additional speech grouping in the inspired model. YAMNet shows the clearest grouping of nature, music and speech, although categories still overlap.

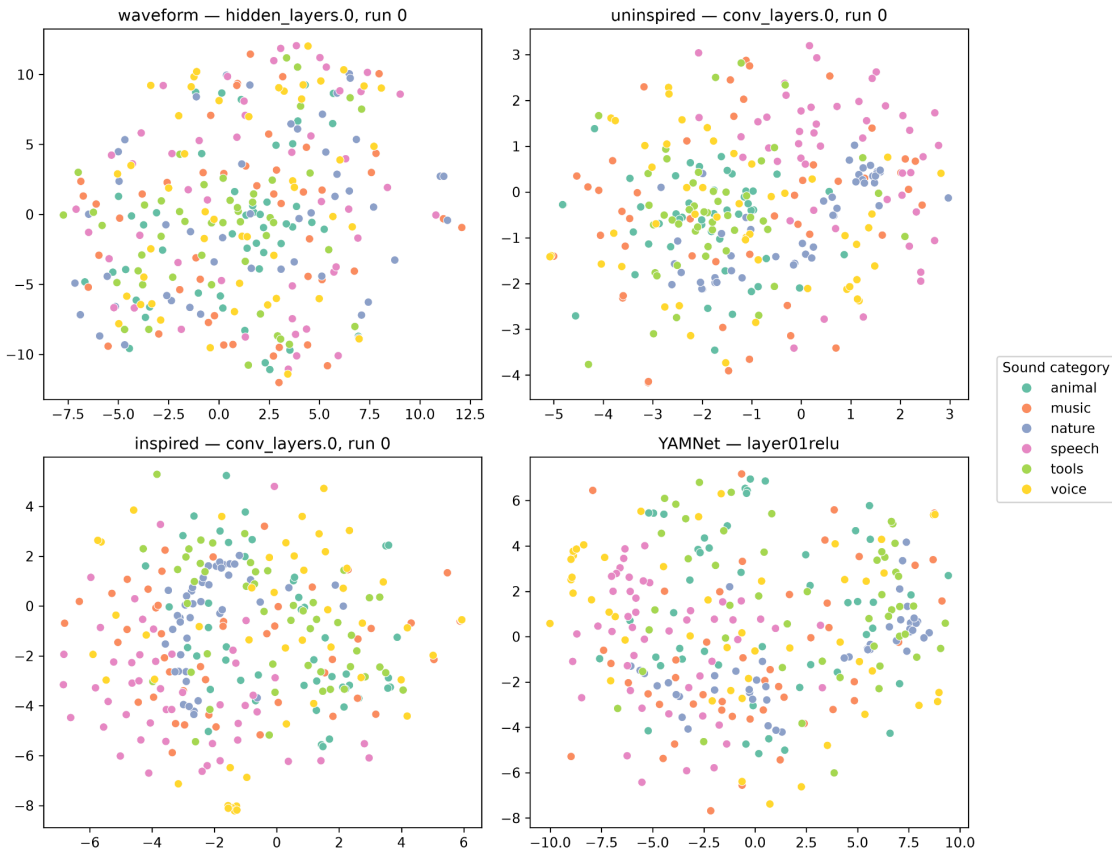


Figure 5. t-SNE visualisations of the first recorded layer of each custom model (trained run 0) and YAMNet. Colours represent the same six categories as Figure 4. Positions and axis values are not directly comparable between panels.

Early layers show more overlap and weaker category separation than the later layers, although some grouping of nature sounds is already visible in the convolutional models and YAMNet. The waveform model remains mostly mixed at both depths.

4. Discussion

Training mainly improved intermediate layers rather than the best overall alignment. This suggests that training develops representations that are more similar to the STG in intermediate layers, while later layers may become more specialised for ESC-50 classification. Architecture had a clearer effect: both spectrogram-based models aligned better with the STG than the waveform model, even though the waveform model had more parameters. This suggests that spectrotemporal processing is more important for brain alignment than model size. The InspiredModel performed best among the custom models, although we cannot separate the effects of the GRU and ReLU6 because both were added together.

Depth also mattered. Alignment increased with depth in the UninspiredModel and InspiredModel, while YAMNet peaked before its final layers. This supports the idea that STG may correspond better to intermediate auditory representations than to final classification outputs. The t-SNE plots support this partly: InspiredModel and YAMNet showed clearer category structure, but category separation alone does not fully explain RSA because RSA compares the overall relationships between sounds, not only visible clusters.

5. Future Work

Future models could separate auditory processing into stages that better reflect known anatomy, rather than simply stacking generic layers. For example, an early tonotopic module could feed into a recurrent intermediate module representing higher auditory processing such as STG. This is motivated by our finding that the strongest improvements in alignment occurred in the convolutional and intermediate layers.

Training could also be changed. Instead of relying only on ESC-50 classification, the model could first use a self-supervised temporal task, such as predicting missing parts of a sound. This may preserve more acoustic and temporal information before classification.

A more direct follow-up would be to include RSA during training as an additional objective, encouraging one intermediate layer to match the STG representational structure while the final layer still performs classification. This would directly test whether preserving a brain-like intermediate representation improves alignment.